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Boyd et al.

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(54) **USER CLASSIFICATION BASED NOTIFICATION**

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(58) **Field of Classification Search**

CPC G06Q 50/01; H04L 51/24; H04L 67/306; G06F 3/0484

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,848,879 B1 9/2014 Coughlan et al.
11,716,243 B2 8/2023 Boyd et al.
2005/0108348 A1 5/2005 Lee
(Continued)

FOREIGN PATENT DOCUMENTS

CN 105592215 A * 5/2016

OTHER PUBLICATIONS

"How to Assign Different Ringtones for Each Contact or Group in Galaxy S3?", FonGuide, Aug. 19, 2012, fonguide.com/2110/how-to-assign-different-ringtones-for-each-contact-or-group-in-galaxy-s3/, accessed Jan. 26, 2024 (Year: 2012).*

(Continued)

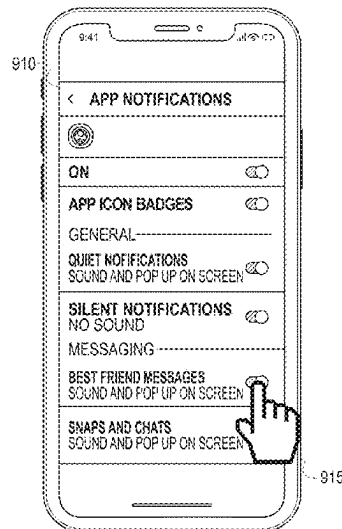
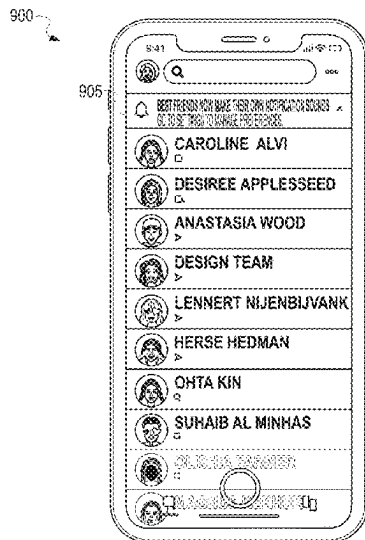
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(57) **ABSTRACT**

A classification based notification system is disclosed which performs operations that include: accessing user profile data associated with a user account associated with a client device, the user profile data including a list of user connections; designating a notification type to a portion of the list of user connections, the notification type corresponding to an alert attribute; receiving a notification from a user connection from among the portion of the list of user connections at the client device; and presenting an alert based on the alert attribute that corresponds with the notification type associated with the portion of the list of user connections at the client device.

17 Claims, 11 Drawing Sheets



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H04L 41/0686 (2022.01)
H04L 51/224 (2022.01)
G06F 3/04847 (2022.01)

(52) **U.S. Cl.**

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 (2013.01); **H04L 41/0686** (2013.01); **H04L**
51/224 (2022.05); **H04L 67/306** (2013.01);
G06F 3/04847 (2013.01)

(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0273689 A1 * 11/2008 Dinoff H04M 19/04
 379/375.01
 2011/0197134 A1 * 8/2011 Kulmala G06Q 50/01
 715/736
 2015/0227611 A1 8/2015 Bao et al.
 2017/0195988 A1 * 7/2017 Tumbi H04W 68/02
 2020/0107176 A1 4/2020 Zavesky et al.
 2021/0377105 A1 12/2021 Boyd et al.

OTHER PUBLICATIONS

“U.S. Appl. No. 16/883,767, Final Office Action mailed Aug. 16, 2022”, 14 pgs.
 “U.S. Appl. No. 16/883,767, Final Office Action mailed Nov. 10, 2022”, 16 pgs.
 “U.S. Appl. No. 16/883,767, Non Final Office Action mailed Feb. 9, 2022”, 14 pgs.
 “U.S. Appl. No. 16/883,767, Non Final Office Action mailed Sep. 29, 2022”, 15 pgs.
 “U.S. Appl. No. 16/883,767, Notice of Allowance mailed Mar. 7, 2023”, 14 pgs.
 “U.S. Appl. No. 16/883,767, Response filed Jan. 9, 2023 to Final Office Action mailed Nov. 10, 2022”, 13 pgs.
 “U.S. Appl. No. 16/883,767, Response filed May 6, 2022 to Non Final Office Action mailed Feb. 9, 2022”, 11 pgs.
 “U.S. Appl. No. 16/883,767, Response filed Aug. 24, 2022 to Final Office Action mailed Aug. 16, 2022”, 10 pgs.
 “U.S. Appl. No. 16/883,767, Response filed Oct. 26, 2022 to Non Final Office Action mailed Sep. 29, 2022”, 12 pgs.
 “How to Assign Different Ringtones for Each Contact or Group in Galaxy S3?”, WEBcazine, <<https://www.webcazine.com/2110/how-to-assign-different-ringtones-for-each-contact-or-group-in-galaxy-s3/>>, (Aug. 19, 2012), 8 pgs.

* cited by examiner

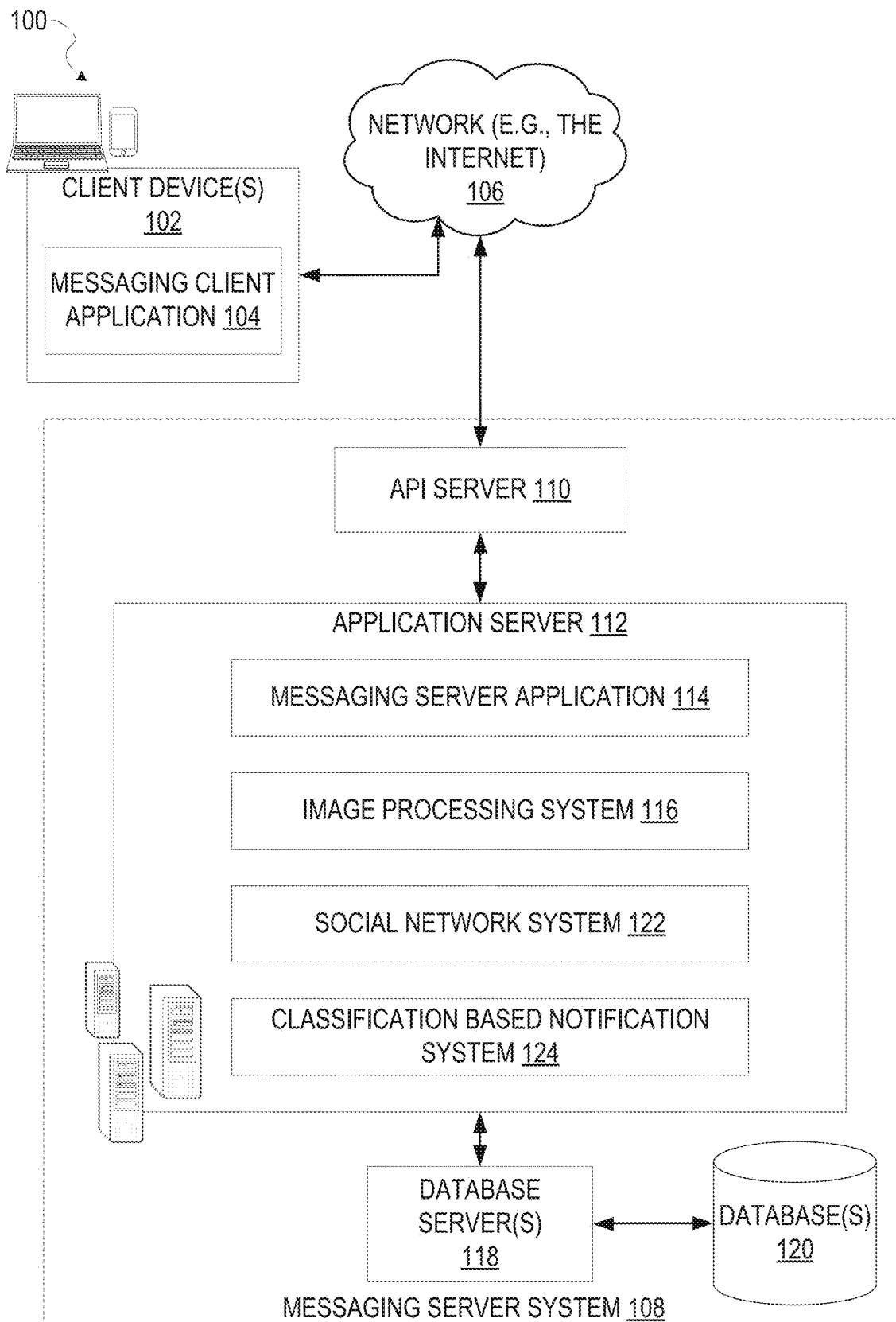


FIG. 1

100

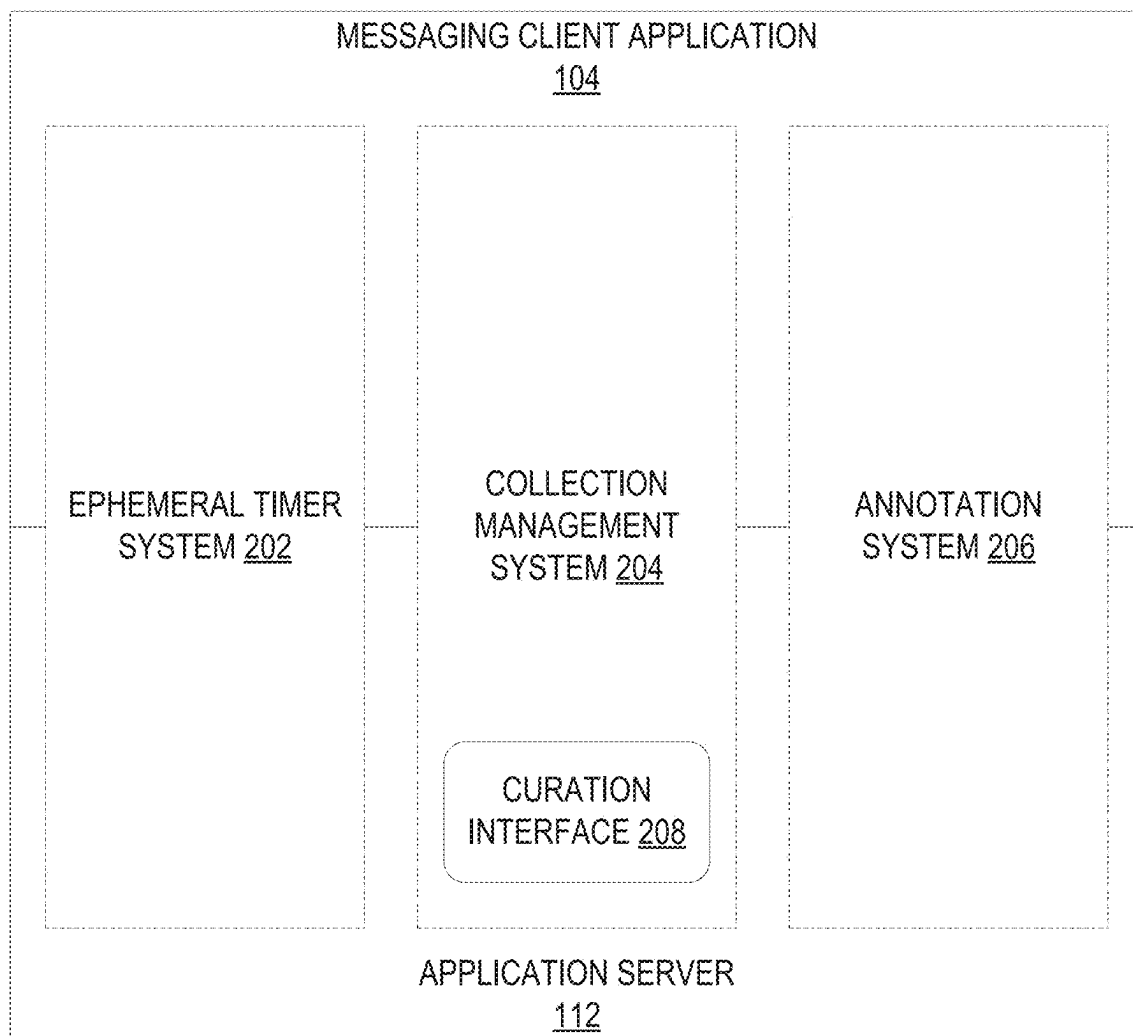


FIG. 2

300

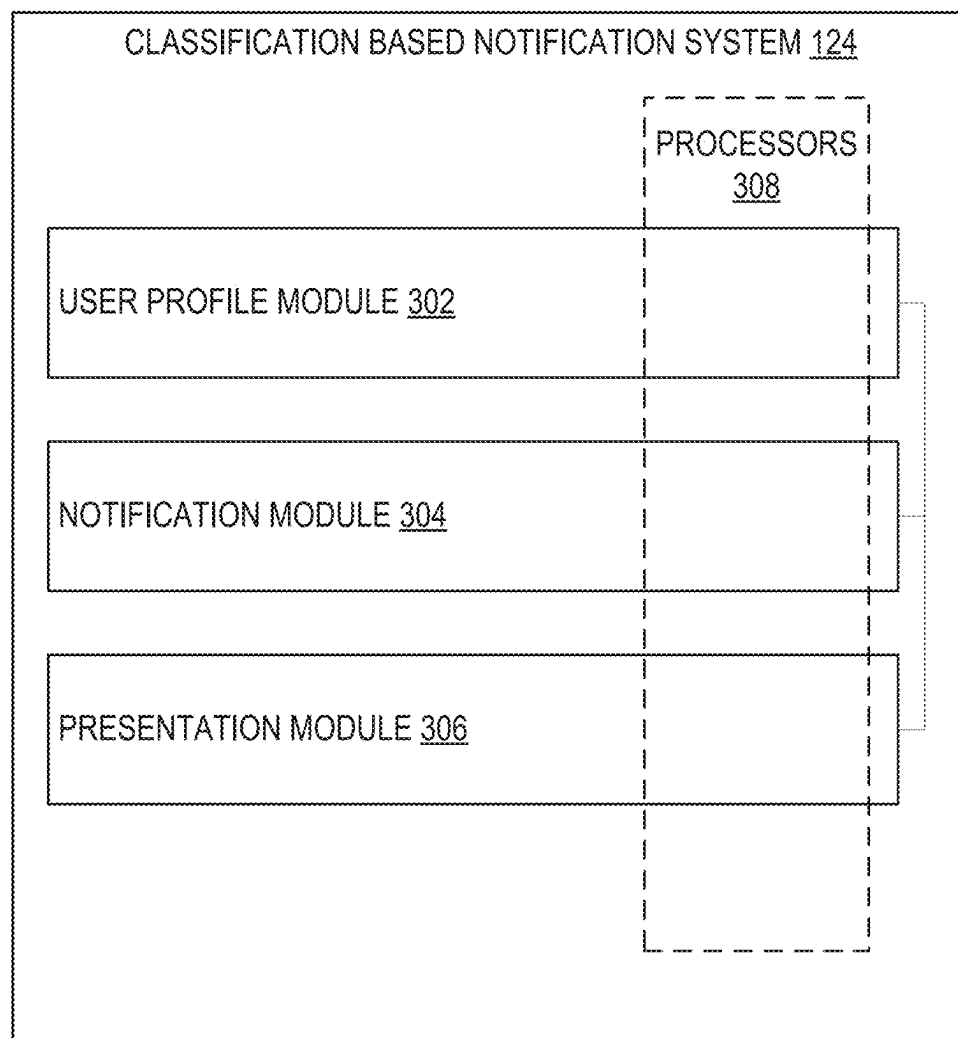



FIG. 3

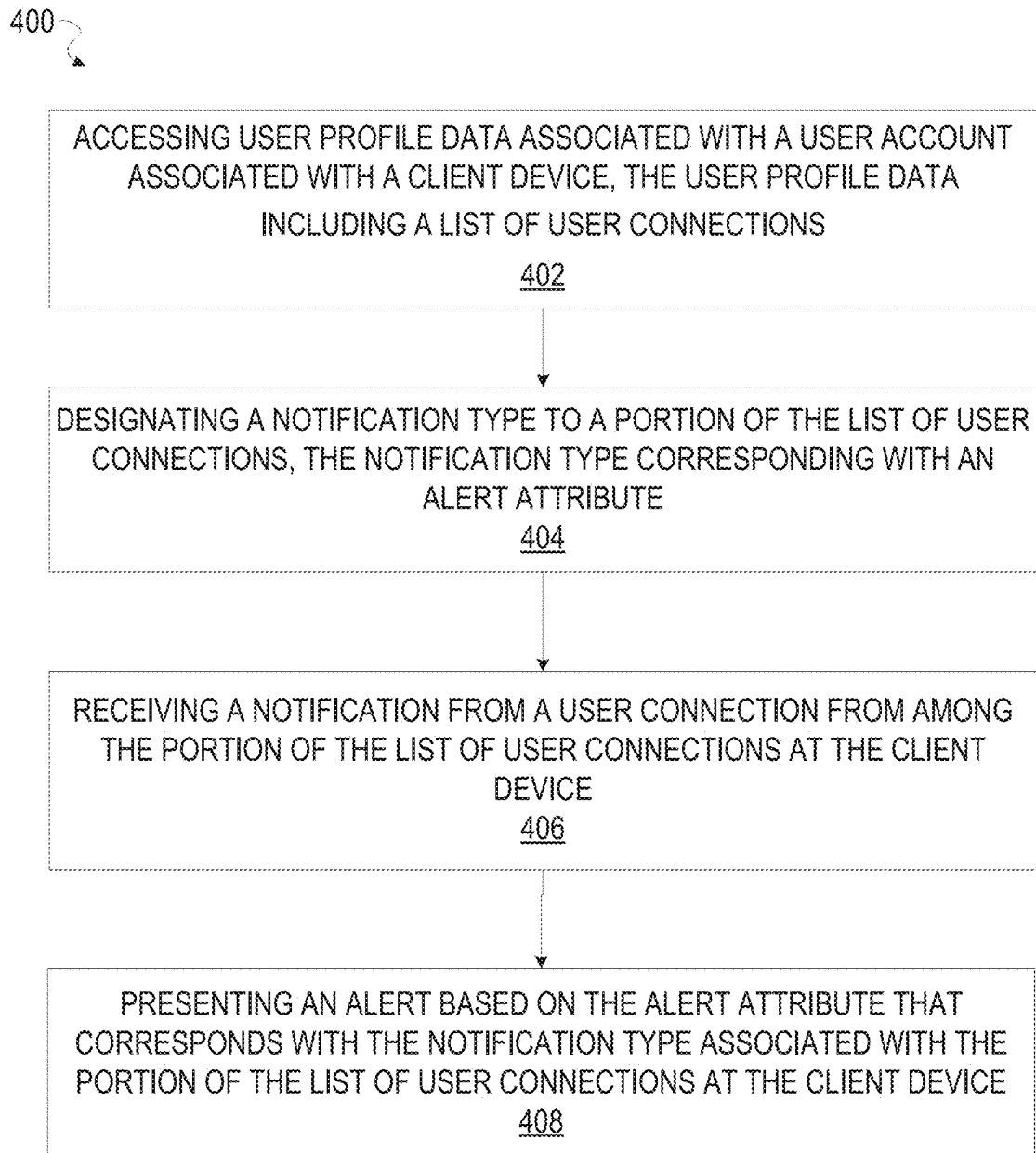
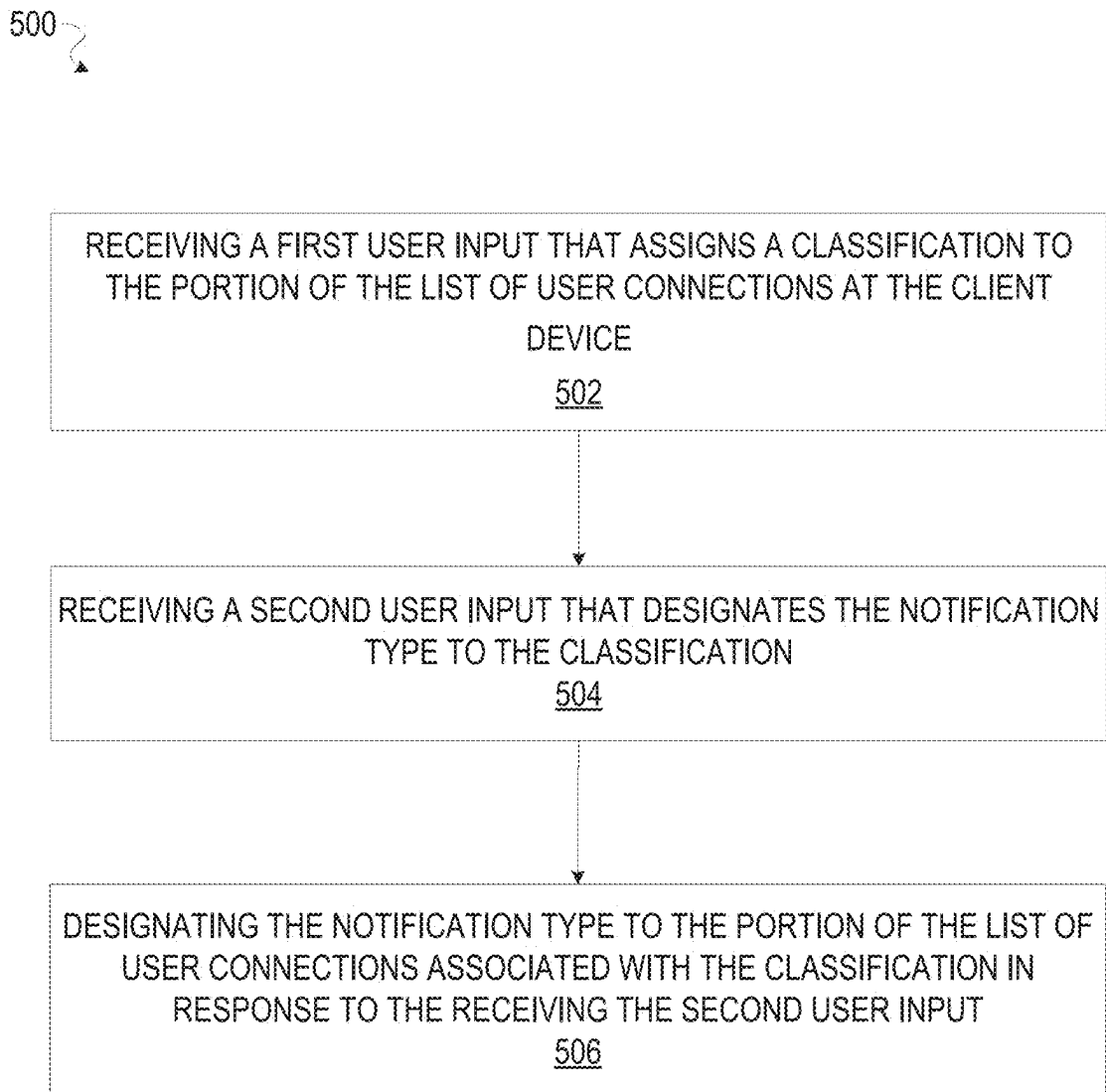


FIG. 4

*FIG. 5*

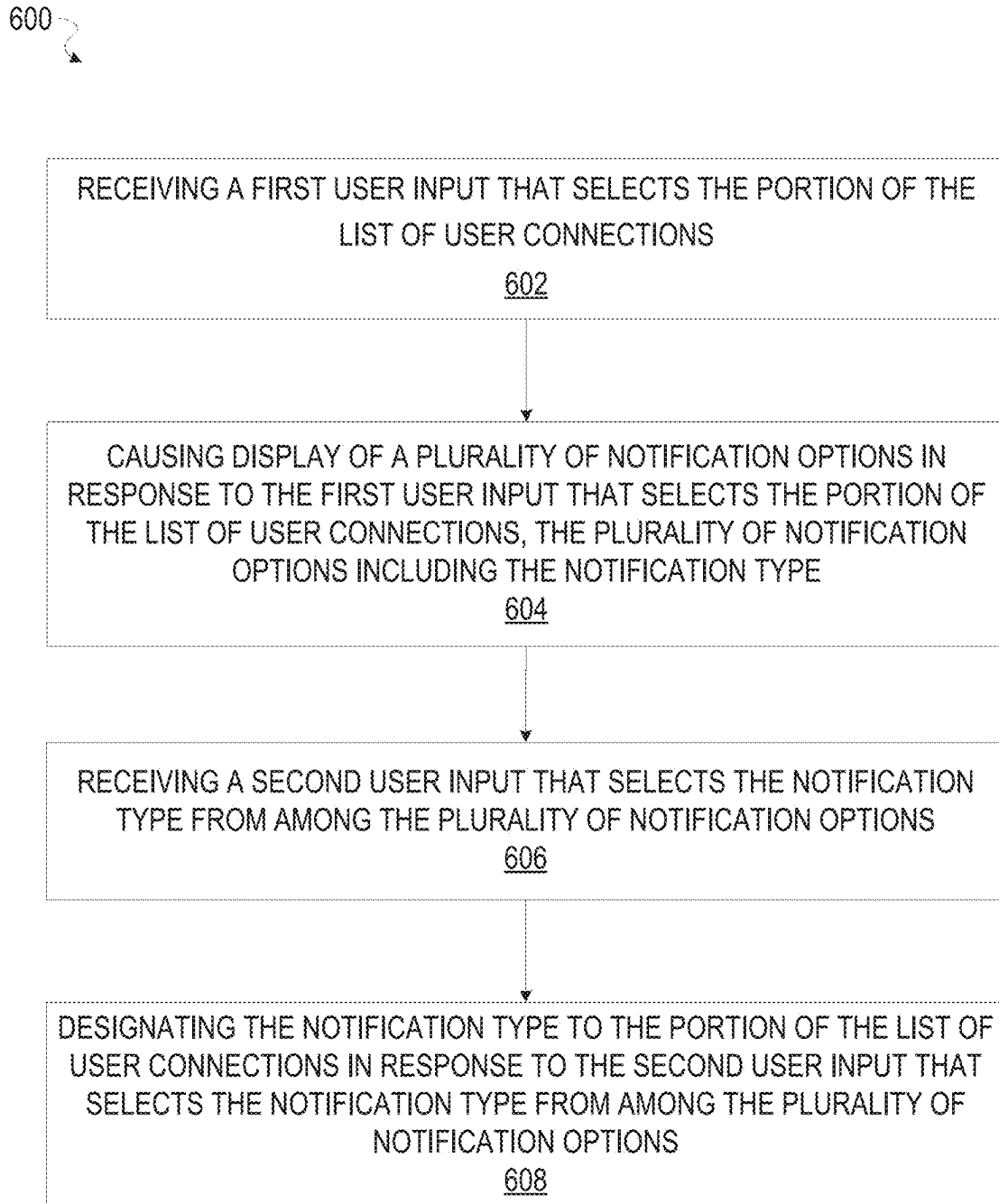


FIG. 6

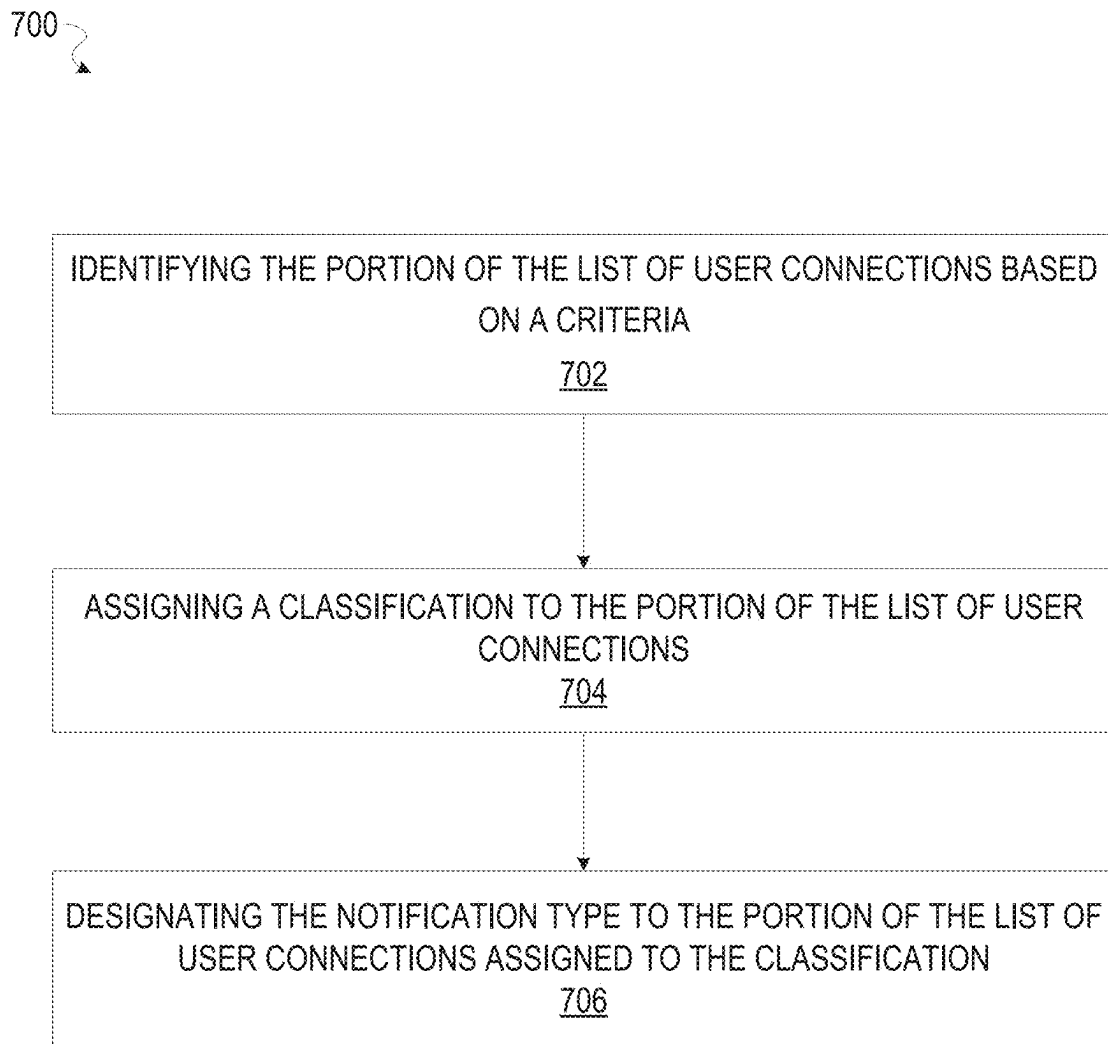


FIG. 7

800

DETERMINING A RATE OF BIDIRECTIONAL COMMUNICATIONS BETWEEN
THE USER ACCOUNT AND EACH USER CONNECTION AMONG THE
PORTION OF THE LIST OF USER CONNECTIONS TRANSGRESSES A
THRESHOLD VALUE

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ASSIGNING THE CLASSIFICATION TO THE PORTION OF THE LIST OF USER
CONNECTIONS IN RESPONSE TO THE DETERMINING THAT THE RATE OF
BIDIRECTIONAL COMMUNICATIONS EXCEEDS THE THRESHOLD VALUE

804*FIG. 8*

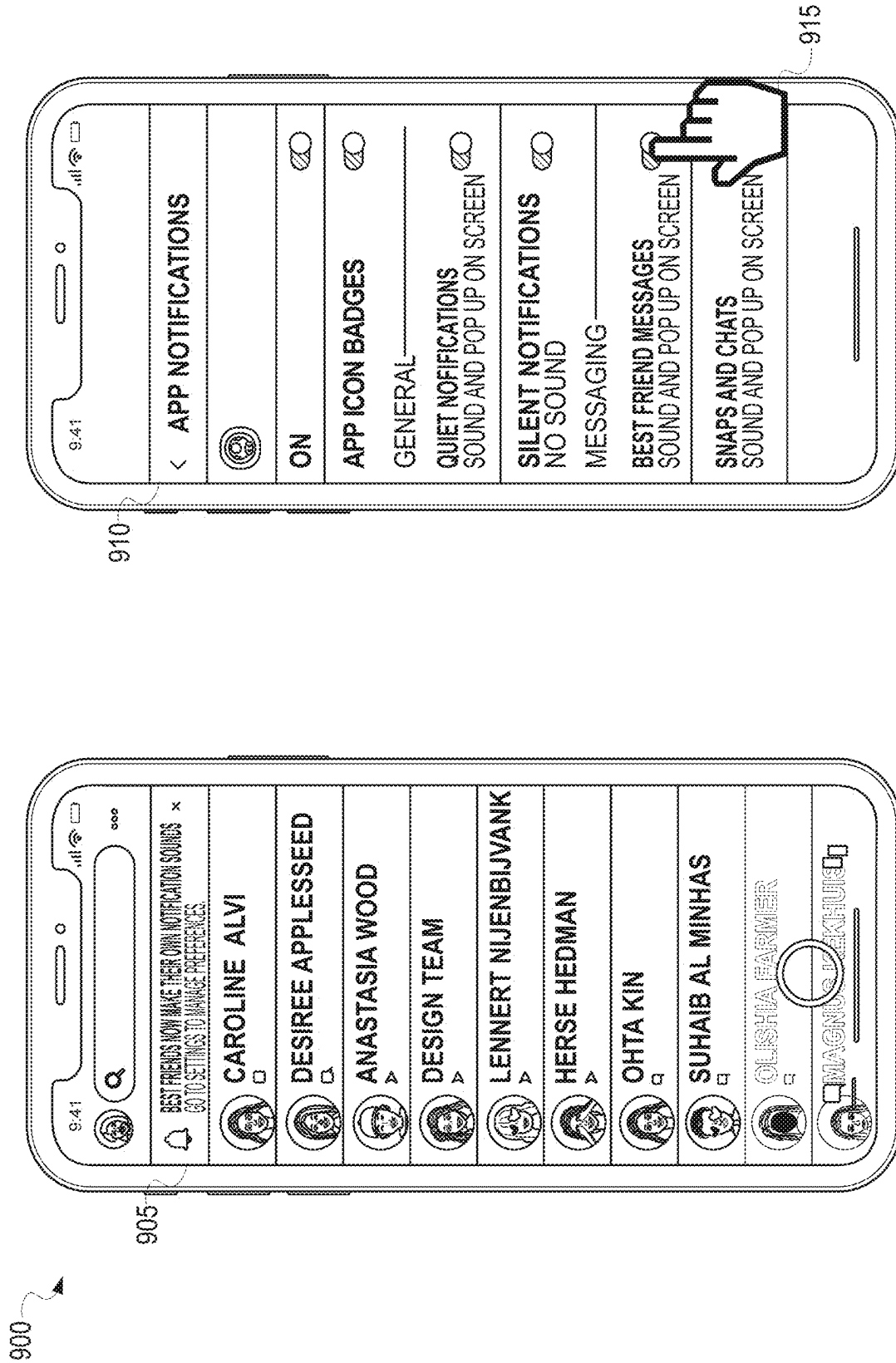


FIG. 9

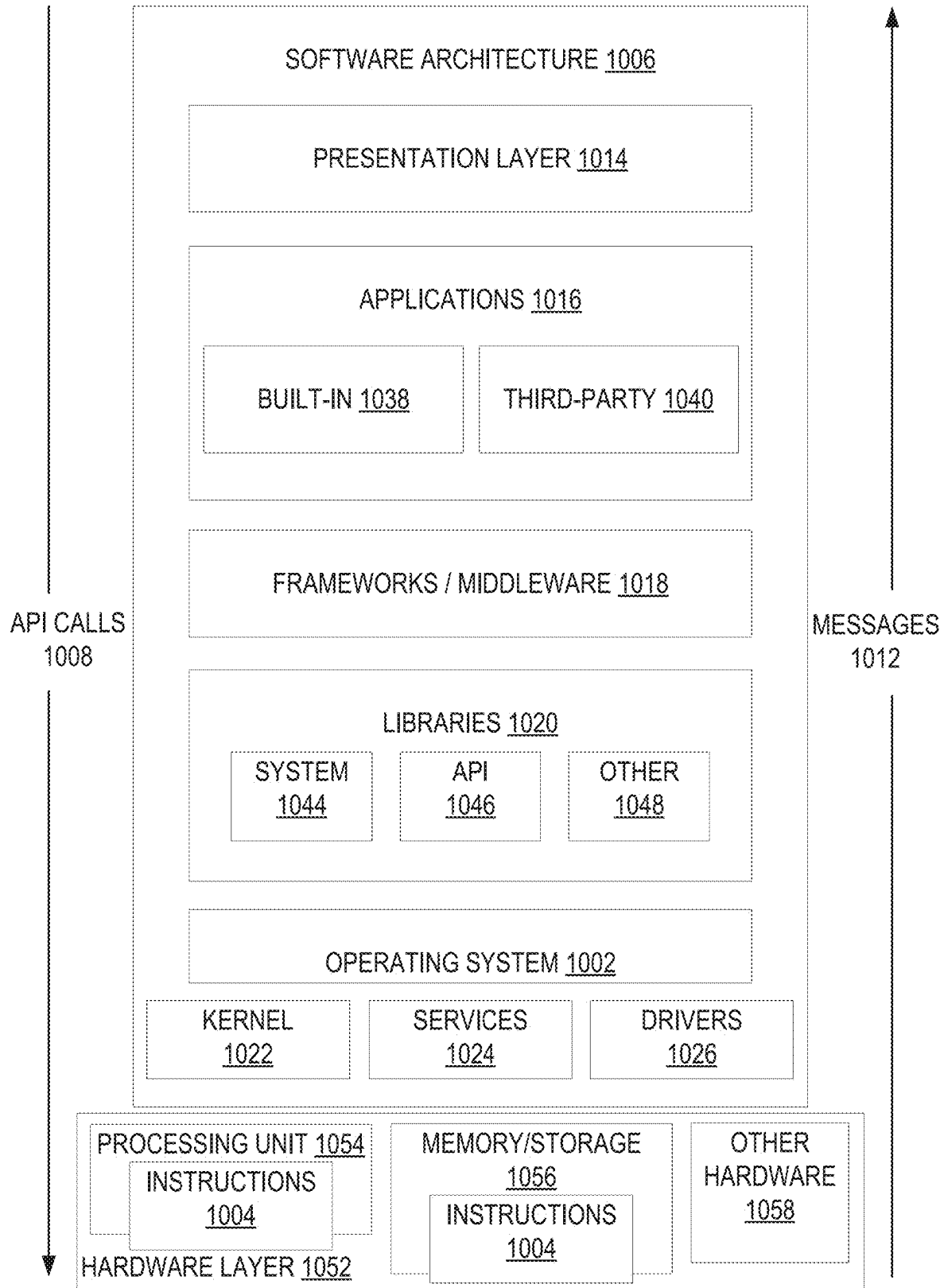


FIG. 10

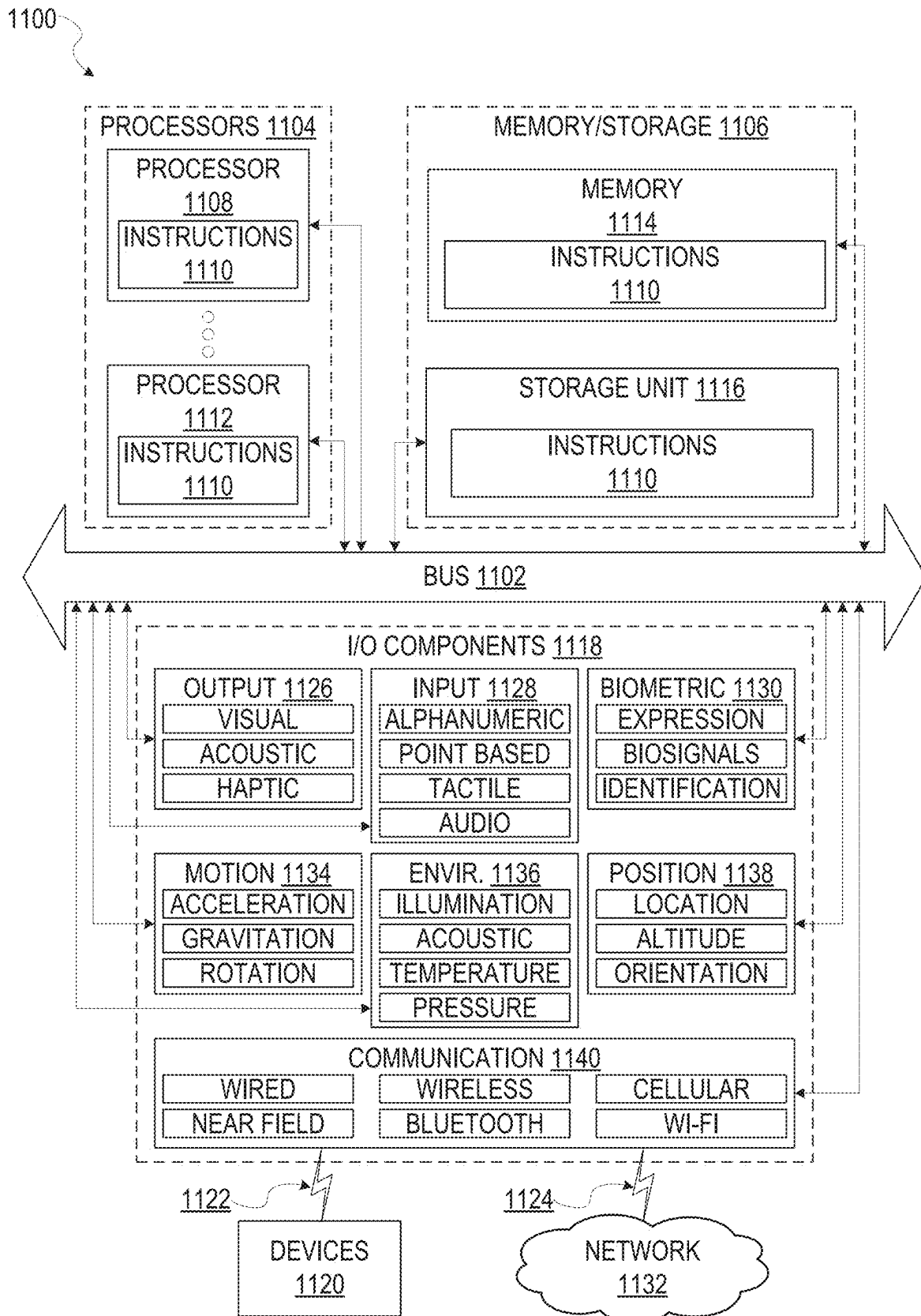


FIG. 11

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USER CLASSIFICATION BASED NOTIFICATION

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 16/883,767, filed May 26, 2020, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

Embodiments of the present disclosure relate generally to mobile computing technology and, more particularly, but not by way of limitation, to systems for presenting notifications.

BACKGROUND

Messaging applications are apps and platforms that enable messaging between users. These messaging applications often enable users to maintain a contact list (i.e., buddy list, user connections, etc.) that comprises a collection of user identifiers. Users of the messaging applications may thereby generate and distribute messages to users from among their contact list, and notifications of the messages may be presented at corresponding client devices.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

FIG. 1 is a block diagram showing an example messaging system for exchanging data (e.g., messages and associated content) over a network in accordance with some embodiments, wherein the messaging system includes a classification based notification system.

FIG. 2 is block diagram illustrating further details regarding a messaging system, according to example embodiments.

FIG. 3 is a block diagram illustrating various modules of a classification based notification system, according to certain example embodiments.

FIG. 4 is a flowchart depicting a method of presenting a notification based on a user classification, according to certain example embodiments.

FIG. 5 is a flowchart depicting a method of presenting a notification based on a user classification, according to certain example embodiments.

FIG. 6 is a flowchart depicting a method of designating a notification type to a user classification, according to certain example embodiments.

FIG. 7 is a flowchart depicting a method of designating a notification type to a user classification, according to certain example embodiments.

FIG. 8 is a flowchart depicting a method of designating a notification type to a user classification, according to certain example embodiments.

FIG. 9 is an interface flow-diagram depicting graphical user interfaces (GUI) generated and displayed by a classification based notification system, according to certain example embodiments.

FIG. 10 is a block diagram illustrating a representative software architecture, which may be used in conjunction

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with various hardware architectures herein described and used to implement various embodiments.

FIG. 11 is a block diagram illustrating components of a machine, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

DETAILED DESCRIPTION

As discussed above, messaging applications provide interfaces for users of the messaging applications to distribute messages between one another. The messaging applications may notify the users of messages by presenting notifications, wherein typically those notifications may comprise a display of message content. While the existing messaging applications are useful, there is always a desire to find better, quicker, and more efficient ways to convey relevant information to users of the messaging applications. Accordingly, in certain example embodiments, a classification based notification system is disclosed which performs operations that include: accessing user profile data associated with a user account associated with a client device, the user profile data including a list of user connections; designating a notification type to a portion of the list of user connections, the notification type corresponding to an alert attribute; receiving a notification from a user connection from among the portion of the list of user connections at the client device; and presenting an alert based on the alert attribute that corresponds with the notification type associated with the portion of the list of user connections at the client device.

The classification based notification system may designate the notification type to the portion of the list of user connections based on an explicit user input received from a user associated with the client device, or in certain embodiments based designate the notification type to the portion of the list of user connections based on attributes associated with the portion of the list of user connections, wherein the attribute includes a classification.

For example, a portion of a list of user connections associated with a user account may be designated one or more classifications based on an input received from a client device, or based on user activities performed by the users among the list of user connections. Accordingly, the classifications may be designating by a user accessing their personal list of user connections, and assigning a classification to one or more of the users among the list of user connections, wherein the classification may indicate a relationship (i.e., family, close friend, co-worker, neighbor, business connection, etc.), or may be based on attributes of communication sessions and user activities performed between the user and each of the users among the list of user connections.

As an illustrative example, in certain embodiments, a frequency or rate in which bidirectional communications (i.e., messages) are exchanged between a user and one or more of the users among the list of user connections may be a factor in determining what classification to assign to a user connection among the list of user connections. For example, the classification based notification system may maintain a repository of user classifications, wherein the user classifications are associated with certain threshold values. Those threshold values may for example define a number of bidirectional communications which must be sent between a user and a user connection within a period of time in order to receive the classification.

In some example embodiments, the alert attribute of the alert may include auditory elements (i.e., a tone, a pattern, a volume), as well as visual elements such as graphical icons, and haptic elements (i.e., vibrate, frequency of vibration, etc.). Accordingly, notifications received from user connections may be presented to a user of a client device based on alert attributes of a corresponding classification. For example, notifications received from a user connection designated as a “close friend” may be presented to the user based on the alert attributes that correspond with the “close friend” classification. These alert attributes may include one or more of a special ring-tone, haptic alert, or graphical icon. The user may therefore quickly determine what type of message or notification they are received based on the attributes of the alert presented, without necessarily having to see the message or notification itself.

FIG. 1 is a block diagram showing an example messaging system 100 for exchanging data (e.g., messages and associated content) over a network. The messaging system 100 includes one or more client device 102 which host a number of applications including a messaging client application 104. Each messaging client application 104 is communicatively coupled to other instances of the messaging client application 104 and a messaging server system 108 via a network 106 (e.g., the Internet).

Accordingly, each messaging client application 104 is able to communicate and exchange data with another messaging client application 104 and with the messaging server system 108 via the network 106. The data exchanged between messaging client applications 104, and between a messaging client application 104 and the messaging server system 108, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

The messaging server system 108 provides server-side functionality via the network 106 to a particular messaging client application 104. While certain functions of the messaging system 100 are described herein as being performed by either a messaging client application 104 or by the messaging server system 108, it will be appreciated that the location of certain functionality either within the messaging client application 104 or the messaging server system 108 is a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system 108, but to later migrate this technology and functionality to the messaging client application 104 where a client device 102 has a sufficient processing capacity.

The messaging server system 108 supports various services and operations that are provided to the messaging client application 104. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client application 104. In some embodiments, this data includes, message content, client device information, geolocation information, media annotation and overlays, message content persistence conditions, social network information, and live event information, as examples. In other embodiments, other data is used. Data exchanges within the messaging system 100 are invoked and controlled through functions available via GUIs of the messaging client application 104.

Turning now specifically to the messaging server system 108, an Application Program Interface (API) server 110 is coupled to, and provides a programmatic interface to, an application server 112. The application server 112 is communicatively coupled to a database server 118, which facili-

tates access to a database 120 in which is stored data associated with messages processed by the application server 112.

Dealing specifically with the Application Program Interface (API) server 110, this server receives and transmits message data (e.g., commands and message payloads) between the client device 102 and the application server 112. Specifically, the Application Program Interface (API) server 110 provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client application 104 in order to invoke functionality of the application server 112. The Application Program Interface (API) server 110 exposes various functions supported by the application server 112, including account registration, login functionality, the sending of messages, via the application server 112, from a particular messaging client application 104 to another messaging client application 104, the sending of media files (e.g., images or video) from a messaging client application 104 to the messaging server application 114, and for possible access by another messaging client application 104, the setting of a collection of media data (e.g., story), the retrieval of a list of friends of a user of a client device 102, the retrieval of such collections, the retrieval of messages and content, the adding and deletion of friends to a social graph, the location of friends within a social graph, opening and application event (e.g., relating to the messaging client application 104).

The application server 112 hosts a number of applications and subsystems, including a messaging server application 114, an image processing system 116, a social network system 122, and a classification based notification system 124. The classification based notification system 124 is configured to generate and present a notification at the client device 110, based on a classification associated with a user connection. Further details of the classification based notification system 124 can be found in FIG. 3 below.

The messaging server application 114 implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client application 104. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available, by the messaging server application 114, to the messaging client application 104. Other processor and memory intensive processing of data may also be performed server-side by the messaging server application 114, in view of the hardware requirements for such processing.

The application server 112 also includes an image processing system 116 that is dedicated to performing various image processing operations, typically with respect to images or video received within the payload of a message at the messaging server application 114.

The social network system 122 supports various social networking functions services, and makes these functions and services available to the messaging server application 114. To this end, the social network system 122 maintains and accesses an entity graph 304 within the database 120. Examples of functions and services supported by the social network system 122 include the identification of other users of the messaging system 100 with which a particular user has relationships or is “following,” and also the identification of other entities and interests of a particular user.

The application server 112 is communicatively coupled to a database server 118, which facilitates access to a database

120 in which is stored data associated with messages processed by the messaging server application 114.

FIG. 2 is block diagram illustrating further details regarding the messaging system 100, according to example embodiments. Specifically, the messaging system 100 is shown to comprise the messaging client application 104 and the application server 112, which in turn embody a number of some subsystems, namely an ephemeral timer system 202, a collection management system 204 and an annotation system 206.

The ephemeral timer system 202 is responsible for enforcing the temporary access to content permitted by the messaging client application 104 and the messaging server application 114. To this end, the ephemeral timer system 202 incorporates a number of timers that, based on duration and display parameters associated with a message, collection of messages (e.g., a story), or graphical element, selectively display and enable access to messages and associated content via the messaging client application 104. Further details regarding the operation of the ephemeral timer system 202 are provided below.

The collection management system 204 is responsible for managing collections of media (e.g., collections of text, image video and audio data). In some examples, a collection of content (e.g., messages, including images, video, text and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system 204 may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client application 104.

The collection management system 204 furthermore includes a curation interface 208 that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface 208 enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system 204 employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain embodiments, compensation may be paid to a user for inclusion of user generated content into a collection. In such cases, the curation interface 208 operates to automatically make payments to such users for the use of their content.

The annotation system 206 provides various functions that enable a user to annotate or otherwise modify or edit media content associated with a message. For example, the annotation system 206 provides functions related to the generation and publishing of media overlays for messages processed by the messaging system 100. The annotation system 206 operatively supplies a media overlay (e.g., a filter, lens) to the messaging client application 104 based on a geolocation of the client device 102. In another example, the annotation system 206 operatively supplies a media overlay to the messaging client application 104 based on other information, such as, social network information of the user of the client device 102. A media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects, as well as animated facial models. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can

be applied to a media content item (e.g., a photo or video) at the client device 102. For example, the media overlay including text that can be overlaid on top of a photograph generated taken by the client device 102. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the annotation system 206 uses the geolocation of the client device 102 to identify a media overlay that includes the name of a merchant at the geolocation of the client device 102. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database 120 and accessed through the database server 118.

In one example embodiment, the annotation system 206 provides a user-based publication platform that enables users to select a geolocation on a map, and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The annotation system 206 generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

In another example embodiment, the annotation system 206 provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the annotation system 206 associates the media overlay of a highest bidding merchant with a corresponding geolocation for a predefined amount of time.

FIG. 3 is a block diagram illustrating components of the classification based notification system 124 that configure the classification based notification system 124 to perform operations to generate and cause display of a notification based on a classification associated with a user connection, according to certain example embodiments.

The classification based notification system 124 is shown as including a user profile module 302, a notification module 304, and a presentation module 306, all configured to communicate with each other (e.g., via a bus, shared memory, or a switch). Any one or more of these modules may be implemented using one or more processors 308 (e.g., by configuring such one or more processors to perform functions described for that module) and hence may include one or more of the processors 308. In certain embodiments, the avatar notification system 124 may include or have access to the database 120, wherein the database 120 may comprise a collection of media content indexed based on user attributes and astrological signs.

Any one or more of the modules described may be implemented using hardware alone (e.g., one or more of the processors 308 of a machine) or a combination of hardware and software. For example, any module described of the avatar notification system 124 may physically include an arrangement of one or more of the processors 308 (e.g., a subset of or among the one or more processors of the machine) configured to perform the operations described herein for that module. As another example, any module of the avatar notification system 124 may include software, hardware, or both, that configure an arrangement of one or more processors 308 (e.g., among the one or more processors of the machine) to perform the operations described herein for that module. Accordingly, different modules of the avatar notification system 124 may include and configure different arrangements of such processors 308 or a single arrangement of such processors 308 at different points in time. Moreover, any two or more modules of the avatar

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notification system **124** may be combined into a single module, and the functions described herein for a single module may be subdivided among multiple modules. Furthermore, according to various example embodiments, modules described herein as being implemented within a single machine, database, or device may be distributed across multiple machines, databases, or devices.

FIG. 4 is a flowchart depicting a method **400** of presenting a notification based on a user classification, according to certain example embodiments. Operations of the method **400** may be performed by the modules described above with respect to FIG. 3. As shown in FIG. 4, the method **400** includes one or more operations **402**, **404**, **406**, and **408**.

At operation **402**, the user profile module **302** accesses user profile data associated with a user account, wherein the user profile data includes a list of user connections (i.e., a buddy list).

At operation **404**, the notification module **304** designates a notification type to a portion of the list of user connections, wherein the notification type corresponds with an alert attribute. In certain example embodiments, the notification module **304** may designate the notification type to the portion of the list of user connections based on one or more of: an explicit user input from the user; based on user attributes of each user from among the portion of the list of user connections; and based on a communication history between the user and each user from among the portion of the list of user connections.

Alert attributes of the notification type may include auditory elements as well as graphical and haptic elements. For example, an alert attribute may include a particular tone or frequency of auditory alert, as well as graphical properties such as one or more graphical icons, or haptic elements such as a vibrate function.

At operation **406**, the client device **102** receives a notification from a user connection from among the portion of the list of user connections. For example, the notification may be a message (i.e., a text message), which may include one or more media objects.

At operation **408**, responsive to receiving the notification from the user connection from among the portion of the list of user connections, the presentation module **306** presents at alert at the client device **102** based on the alert attribute that corresponds with the notification type associated with the portion of the list of user connections. For example, the alert may comprise a particular auditory alert associated with the portion of the list of user connections.

FIG. 5 is a flowchart depicting a method **500** of presenting a notification based on a user classification, according to certain example embodiments. Operations of the method **500** may be performed by the modules described above with respect to FIG. 3. As shown in FIG. 5, the method **500** includes one or more operations **502**, **504**, and **506**.

Operations of the method **500** depicted in FIG. 5 provide an embodiment for designating a notification type from among a plurality of notification types to a portion of a list of user connections associated with a user account, wherein the portion of the list of user connections may comprise one or more user identifiers.

At operation **502**, the user profile module **302** receives a first user input that assigns a classification to the portion of the list of user connections at the client device **102**. For example, a user associated with the user account may provide an input that selects or otherwise identifies one or more user identifiers that comprise the portion of the list of user identifiers.

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According to certain embodiments, the classification may include an identification of a relationship or relationship type. For example, the classification may include: co-worker; work connection; family; close friend; business connection; neighbor; class-mate; or a user defined classification. The user of the client device **102** may provide an input that assigns the classification to each user among the portion of the list of user connections.

At operation **504**, the user profile module **302** receives a second user input that designates a notification type to the classification. As discussed above, the notification type may correspond with a particular set of alert attributes, such as auditory elements, haptic elements, and graphical elements.

At operation **506**, responsive to receiving the second user input at operation **504**, the user profile module **302** designates the notification type to the portion of the list of user connections that correspond with the classification. Accordingly, messages and notification receives at the client device **102** from the portion of the list of user connections may be presented to a user of the client device **102** based on the alert attributes of the corresponding notification type.

FIG. 6 is a flowchart depicting a method **600** of designating a notification type to a user classification, according to certain example embodiments. Operations of the method **600** may be performed by the modules described above with respect to FIG. 3. As shown in FIG. 6, the method **600** includes one or more operations **602**, **604**, **606**, and **608**.

At operation **602**, the user profile module **302** receives a first input that selects a portion of a list of user connections associated with a user account from a client device **102**. For example, a user of the client device **102** may provide a plurality of inputs that explicitly identify each user from among the portion of the list of user connections (i.e., based on corresponding user identifiers). In further embodiments, the user of the client device **102** may provide an input that identifies the portion of the list of user connections based on a user attribute associated with each user among the portion of the list of user connections.

At operation **604**, responsive to the user profile module **302** receiving the input that selects the portion of the list of user connections, the presentation module **306** causes display of a plurality of notification options, wherein the plurality of notification options comprise notification types that have corresponding alert attributes.

At operation **606**, the user profile module **302** receives a second user input that selects a notification type from among the plurality of notification options. For example, the second user input may comprise a tactile input that selects a graphical icon that corresponds with the notification type.

At operation **608**, responsive to receiving the second user input that selects the notification type from among the plurality of notification options, the user profile module **302** designates the notification type to the portion of the list of user connections.

FIG. 7 is a flowchart depicting a method **700** of designating a notification type to a user classification, according to certain example embodiments. Operations of the method **700** may be performed by the modules described above with respect to FIG. 3. As shown in FIG. 7, the method **700** includes one or more operations **702**, **704**, and **706**.

At operation **702**, the user profile module **302** identifies the portion of the list of user connections based on a criteria. For example, the criteria may include user attributes, location data, as well as a communication history between a user of the client device **102** and each user among the portion of the list of user connections.

At operation **704**, the user profile module **302** assigns a user classification to each user among the portion of the list of user connections responsive to identifying the portion of the list of user connections based on the criteria. The classification may include the classifications discussed above with respect to the methods **400** and **500** depicted in FIGS. **4** and **5**.

At operation **706**, the notification module **306** designates a notification type to the portion of the list of user connections based on the classification that corresponds with the portion of the list of user connections, in response to identifying the portion of the list of user connections based on the criteria.

FIG. **8** is a flowchart depicting a method **800** of designating a notification type to a user classification, according to certain example embodiments. Operations of the method **800** may be performed by the modules described above with respect to FIG. **3**. As shown in FIG. **8**, the method **800** includes one or more operations **802**, and **804**. According to certain embodiments, the method **800** may be performed as a subroutine of the method **700** depicted in FIG. **7**.

In certain example embodiments, the criteria described in the method **700** may comprise a threshold value, such as a threshold number of bidirectional messages sent between a user of the client device **102** and each of the user connections from among the portion of the list of user connections, over a period of time.

At operation **802**, the user profile module **302** determines a rate of bidirectional communication between a user associated with a user account, and each user among a list of user connections. For example, the user profile module **302** may access a user account of the user, wherein the user account comprises a communication log of all messages sent between the user and user connections of the user.

At operation **804**, responsive to determining that each user connection from among the portion of the list of user connections are associated with a communication history wherein at least the threshold number of bidirectional communications have been sent, the user profile module **302** assigns a classification to the portion of the list of user connections, wherein the classification corresponds with a notification type that includes alert attributes.

FIG. **9** is an interface flow-diagram **900** depicting graphical user interfaces (GUI) generated and displayed by a classification based notification system **124**, according to certain example embodiments.

As seen in the GUI **905**, the classification based notification system **124** may present a notification to request a user of a client device **102** to provide inputs to define notification parameters of notifications received from different user classifications. For example, the classification based notification system **124** may classify user connections from among a list of user connections based on a communication history between a user of the client device **102** and each user connections from among the list of user connections. The classification may for example include a “best friend” classification, wherein the best friend classification may be assigned to a portion of the list of user connections with which the user of the client device **102** communicates with the most. For example, the top 7 user connections in which the user of the client device **102** communicates with.

In further embodiments, and as discussed in the method **800** depicted in FIG. **8**, the classification based notification system **124** may designate the best friend classification to user connections in which a communication history indicates that the user of the client device **102** engages in a threshold number of bidirectional communications (i.e.,

messages). For example, user connections in which the user of the client device **102** sends at least 10 messages to per day may be assigned the best friend classification.

As seen in the GUI **910**, a user of the client device **102** may provide an input to define one or more notification parameters associated with the notifications presented when a user connection classified as a best friend send them a message. For example, the user of the client device **102** may provide an input **915** that enables or disables the notification type associated with the best friend classification.

Responsive to receiving the input that enables the notification type associated with the best friend classification, the classification based notification system **124** may identify the portion of the list of user connections based on the methods described herein. For example, the classification based notification system **124** may identify the users based on the bidirectional communication history, or in further embodiments based on user attributes of the user from among the portion of the list of user connections, or explicit user inputs received from the user of the client device **102**, wherein the user input assign the best friend designation to the portion of the list of user connections.

The classification based notification system **124** may assign the notification type to the portion of the list of user connections responsive to the input **915**. Subsequent to receiving the input **915** that enables the features, messages received from the portion of the list of user connections may be presented based on the alert attributes of the notification type that corresponds with the classification.

Software Architecture

FIG. **10** is a block diagram illustrating an example software architecture **1006**, which may be used in conjunction with various hardware architectures herein described. FIG. **10** is a non-limiting example of a software architecture and it will be appreciated that many other architectures may be implemented to facilitate the functionality described herein. The software architecture **1006** may execute on hardware such as machine **1100** of FIG. **11** that includes, among other things, processors **1104**, memory **1114**, and I/O components **1118**. A representative hardware layer **1052** is illustrated and can represent, for example, the machine **1000** of FIG. **10**. The representative hardware layer **1052** includes a processing unit **1054** having associated executable instructions **1004**. Executable instructions **1004** represent the executable instructions of the software architecture **1006**, including implementation of the methods, components and so forth described herein. The hardware layer **1052** also includes memory and/or storage modules memory/storage **1056**, which also have executable instructions **1004**. The hardware layer **1052** may also comprise other hardware **1058**.

In the example architecture of FIG. **10**, the software architecture **1006** may be conceptualized as a stack of layers where each layer provides particular functionality. For example, the software architecture **1006** may include layers such as an operating system **1002**, libraries **1020**, applications **1016** and a presentation layer **1014**. Operationally, the applications **1016** and/or other components within the layers may invoke application programming interface (API) API calls **1008** through the software stack and receive a response as in response to the API calls **1008**. The layers illustrated are representative in nature and not all software architectures have all layers. For example, some mobile or special purpose operating systems may not provide a frameworks/

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middleware **1018**, while others may provide such a layer. Other software architectures may include additional or different layers.

The operating system **1002** may manage hardware resources and provide common services. The operating system **1002** may include, for example, a kernel **1022**, services **1024** and drivers **1026**. The kernel **1022** may act as an abstraction layer between the hardware and the other software layers. For example, the kernel **1022** may be responsible for memory management, processor management (e.g., scheduling), component management, networking, security settings, and so on. The services **1024** may provide other common services for the other software layers. The drivers **1026** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1026** include display drivers, camera drivers, Bluetooth® drivers, flash memory drivers, serial communication drivers (Universal Serial Bus (USB) drivers), Wi-Fi® drivers, audio drivers, power management drivers, and so forth depending on the hardware configuration.

The libraries **1020** provide a common infrastructure that is used by the applications **1016** and/or other components and/or layers. The libraries **1020** provide functionality that allows other software components to perform tasks in an easier fashion than to interface directly with the underlying operating system **1002** functionality (e.g., kernel **1022**, services **1024** and/or drivers **1026**). The libraries **1020** may include system libraries **1044** (e.g., C standard library) that may provide functions such as memory allocation functions, string manipulation functions, mathematical functions, and the like. In addition, the libraries **1020** may include API libraries **1046** such as media libraries (e.g., libraries to support presentation and manipulation of various media format such as MPREG4, H.264, MP3, AAC, AMR, JPG, PNG), graphics libraries (e.g., an OpenGL framework that may be used to render 2D and 3D in a graphic content on a display), database libraries (e.g., SQLite that may provide various relational database functions), web libraries (e.g., WebKit that may provide web browsing functionality), and the like. The libraries **1020** may also include a wide variety of other libraries **1048** to provide many other APIs to the applications **1016** and other software components/modules.

The frameworks/middleware **1018** (also sometimes referred to as middleware) provide a higher-level common infrastructure that may be used by the applications **1016** and/or other software components/modules. For example, the frameworks/middleware **1018** may provide various graphic user interface (GUI) functions, high-level resource management, high-level location services, and so forth. The frameworks/middleware **1018** may provide a broad spectrum of other APIs that may be utilized by the applications **1016** and/or other software components/modules, some of which may be specific to a particular operating system **1002** or platform.

The applications **1016** include built-in applications **1038** and/or third-party applications **1040**. Examples of representative built-in applications **1038** may include, but are not limited to, a contacts application, a browser application, a book reader application, a location application, a media application, a messaging application, and/or a game application. Third-party applications **1040** may include an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform, and may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or other mobile operating systems. The third-party applications **1040** may invoke the API calls

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1008 provided by the mobile operating system (such as operating system **1002**) to facilitate functionality described herein.

The applications **1016** may use built in operating system functions (e.g., kernel **1022**, services **1024** and/or drivers **1026**), libraries **1020**, and frameworks/middleware **1018** to create user interfaces to interact with users of the system. Alternatively, or additionally, in some systems interactions with a user may occur through a presentation layer, such as presentation layer **1014**. In these systems, the application/component “logic” can be separated from the aspects of the application/component that interact with a user.

FIG. **11** is a block diagram illustrating components of a machine **1100**, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **11** shows a diagrammatic representation of the machine **1100** in the example form of a computer system, within which instructions **1110** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1100** to perform any one or more of the methodologies discussed herein may be executed. As such, the instructions **1110** may be used to implement modules or components described herein. The instructions **1110** transform the general, non-programmed machine **1100** into a particular machine **1100** programmed to carry out the described and illustrated functions in the manner described. In alternative embodiments, the machine **1100** operates as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1100** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1100** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1110**, sequentially or otherwise, that specify actions to be taken by machine **1100**. Further, while only a single machine **1100** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1110** to perform any one or more of the methodologies discussed herein.

The machine **1100** may include processors **1104**, memory **1106**, and I/O components **1118**, which may be configured to communicate with each other such as via a bus **1102**. The memory/storage **1106** may include a memory **1114**, such as a main memory, or other memory storage, and a storage unit **1116**, both accessible to the processors **1104** such as via the bus **1102**. The storage unit **1116** and memory **1114** store the instructions **1110** embodying any one or more of the methodologies or functions described herein. The instructions **1110** may also reside, completely or partially, within the memory **1114**, within the storage unit **1116**, within at least one of the processors **1104** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1100**. Accordingly, the memory **1114**, the storage unit **1116**, and the memory of processors **1104** are examples of machine-readable media.

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The I/O components **1118** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1118** that are included in a particular machine **1100** will depend on the type of machine. For example, portable machines such as mobile phones will likely include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1118** may include many other components that are not shown in FIG. **11**. The I/O components **1118** are grouped according to functionality merely for simplifying the following discussion and the grouping is in no way limiting. In various example embodiments, the I/O components **1118** may include output components **1126** and input components **1128**. The output components **1126** may include visual components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input components **1128** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

In further example embodiments, the I/O components **1118** may include biometric components **1130**, motion components **1134**, environmental environment components **1136**, or position components **1138** among a wide array of other components. For example, the biometric components **1130** may include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components **1134** may include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The environment components **1136** may include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometer that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detect concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **1138** may include location sensor components (e.g., a Global Position system (GPS) receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

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Communication may be implemented using a wide variety of technologies. The I/O components **1118** may include communication components **1140** operable to couple the machine **1100** to a network **1132** or devices **1120** via coupling **1122** and coupling **1124** respectively. For example, the communication components **1140** may include a network interface component or other suitable device to interface with the network **1132**. In further examples, communication components **1140** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1120** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a Universal Serial Bus (USB)).

Moreover, the communication components **1140** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1140** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Data-glyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1140**, such as, location via Internet Protocol (IP) geo-location, location via Wi-Fi® signal triangulation, location via detecting a NFC beacon signal that may indicate a particular location, and so forth.

Glossary

“CARRIER SIGNAL” in this context refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible medium to facilitate communication of such instructions. Instructions may be transmitted or received over the network using a transmission medium via a network interface device and using any one of a number of well-known transfer protocols.

“CLIENT DEVICE” in this context refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smart phones, tablets, ultra books, notebooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

“COMMUNICATIONS NETWORK” in this context refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of net-

work, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other type of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard setting organizations, other long range protocols, or other data transfer technology.

“EMPHEMERAL MESSAGE” in this context refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

“MACHINE-READABLE MEDIUM” in this context refers to a component, device or other tangible media able to store instructions and data temporarily or permanently and may include, but is not be limited to, random-access memory (RAM), read-only memory (ROM), buffer memory, flash memory, optical media, magnetic media, cache memory, other types of storage (e.g., Erasable Programmable Read-Only Memory (EEPROM)) and/or any suitable combination thereof. The term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions. The term “machine-readable medium” shall also be taken to include any medium, or combination of multiple media, that is capable of storing instructions (e.g., code) for execution by a machine, such that the instructions, when executed by one or more processors of the machine, cause the machine to perform any one or more of the methodologies described herein. Accordingly, a “machine-readable medium” refers to a single storage apparatus or device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” excludes signals per se.

“COMPONENT” in this context refers to a device, physical entity or logic having boundaries defined by function or subroutine calls, branch points, application program interfaces (APIs), or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various example embodiments, one or more computer systems (e.g., a standalone computer system,

a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a Field-Programmable Gate Array (FPGA) or an Application Specific Integrated Circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In embodiments in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at

least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an Application Program Interface (API)). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the processors or processor-implemented components may be distributed across a number of geographic locations.

“PROCESSOR” in this context refers to any circuit or virtual circuit (a physical circuit emulated by logic executing on an actual processor) that manipulates data values according to control signals (e.g., “commands”, “op codes”, “machine code”, etc.) and which produces corresponding output signals that are applied to operate a machine. A processor may, for example, be a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) processor, a Complex Instruction Set Computing (CISC) processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC) or any combination thereof. A processor may further be a multi-core processor having two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously.

“TIMESTAMP” in this context refers to a sequence of characters or encoded information identifying when a certain event occurred, for example giving date and time of day, sometimes accurate to a small fraction of a second.

“LIFT” in this context is a measure of the performance of a targeted model at predicting or classifying cases as having an enhanced response (with respect to a population as a whole), measured against a random choice targeting model.

“PHONEME ALIGNMENT” in this context, a phoneme is a unit of speech that differentiates one word from another. One phoneme may consist of a sequence of closure, burst, and aspiration events; or, a diphthong may transition from a back vowel to a front vowel. A speech signal may therefore be described not only by what phonemes it contains, but also the locations of the phonemes. Phoneme alignment may therefore be described as a “time-alignment” of phonemes in a waveform, in order to determine an appropriate sequence and location of each phoneme in a speech signal.

“AUDIO-TO-VISUAL CONVERSION” in this context refers to the conversion of audible speech signals into visible

speech, wherein the visible speech may include a mouth shape representative of the audible speech signal.

“TIME DELAYED NEURAL NETWORK (TDINN)” in this context, a TDNN is an artificial neural network architecture whose primary purpose is to work on sequential data. An example would be converting continuous audio into a stream of classified phoneme labels for speech recognition.

“BI-DIRECTIONAL LONG-SHORT TERM MEMORY (BLSTM)” in this context refers to a recurrent neural network (RNN) architecture that remembers values over arbitrary intervals. Stored values are not modified as learning proceeds. RNNs allow forward and backward connections between neurons. BLSTM are well-suited for the classification, processing, and prediction of time series, given time lags of unknown size and duration between events.

What is claimed is:

1. A method comprising:

receiving, from a client device, an input that designates a notification type to a user attribute that includes a rate of communication that comprises a minimum number of bidirectional communications within a predefined time period in order to be designated with the notification type, the notification type comprising one or more alert attributes;

designating the notification type to a portion of a list of user connections based on the rate of communication, the portion of the list of user connections including a user account that includes a communication history that indicates the rate of communication designated by the input;

receiving a notification from the user account; and presenting an alert at the client device based on the notification type designated to the user attribute by the input.

2. The method of claim 1, wherein the one or more alert attributes include one or more of the list comprising:

a tone from among one or more tones;
a volume setting from among a set of volume settings;
a haptic element from among one or more haptic elements; and
a graphical icon from among one or more graphical icons.

3. The method of claim 1, wherein the user attribute includes one or more of the list comprising:

a user identifier from among one or more user identifiers;
location data; and
a user action from among one or more user actions.

4. The method of claim 1, wherein the designating the notification type to the portion of the list of user connections based on the user attribute includes:

identifying the portion of the list of user connections based on the user attribute.

5. The method of claim 1, wherein the notification includes a message received from the user account.

6. The method of claim 1, wherein the user attribute includes a classification, and the designating the notification type to the portion of the list of user connections includes: receiving a request to assign the classification to the portion of the list of user connections.

7. A system comprising:

a memory; and
at least one hardware processor coupled to the memory and comprising instructions that causes the system to perform operations comprising:

receiving, from a client device, an input that designates a notification type to a user attribute that includes a rate of communication that comprises a minimum number

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of bidirectional communications within a predefined time period in order to be designated with the notification type, the notification type comprising one or more alert attributes;

designating the notification type to a portion of a list of user connections based on the rate of communication, the portion of the list of user connections including a user account that includes a communication history that indicates the rate of communication designated by the input;

receiving a notification from the user account; and presenting an alert at the client device based on the notification type designated to the user attribute by the input.

8. The system of claim 7, wherein the one or more alert attributes include one or more of the list comprising:

a tone from among one or more tones;
a volume setting from among a set of volume settings;
a haptic element from among one or more haptic elements; and
a graphical icon from among one or more graphical icons.

9. The system of claim 7, wherein the user attribute includes one or more of the list comprising:

a user identifier from among one or more user identifiers; location data; and
a user action from among one or more user actions.

10. The system of claim 7, wherein the designating the notification type to the portion of the list of user connections based on the user attribute includes:

identifying the portion of the list of user connections based on the user attribute.

11. The system of claim 7, wherein the notification includes a message received from the user account.

12. The system of claim 7, wherein the user attribute includes a classification, and the designating the notification type to the portion of the list of user connections includes: receiving a request to assign the classification to the portion of the list of user connections.

13. A non-transitory machine-readable storage medium comprising instructions that, when executed by one or more processors of a machine, cause the machine to perform operations comprising:

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receiving, from a client device, an input that designates a notification type to a user attribute that includes a rate of communication that comprises a minimum number of bidirectional communications within a predefined time period in order to be designated with the notification type, the notification type comprising one or more alert attributes;

designating the notification type to a portion of a list of user connections based on the rate of communication, the portion of the list of user connections including a user account that includes a communication history that indicates the rate of communication designated by the input;

receiving a notification from the user account; and presenting an alert at the client device based on the notification type designated to the user attribute by the input.

14. The non-transitory machine-readable storage medium of claim 13, wherein the one or more alert attributes include one or more of the list comprising:

a tone from among one or more tones;
a volume setting from among a set of volume settings;
a haptic element from among one or more haptic elements; and
a graphical icon from among one or more graphical icons.

15. The non-transitory machine-readable storage medium of claim 13, wherein the user attribute includes one or more of the list comprising:

a user identifier from among one or more user identifiers; location data; and
a user action from among one or more user actions.

16. The non-transitory machine-readable storage medium of claim 13, wherein the designating the notification type to the portion of the list of user connections based on the user attribute includes:

identifying the portion of the list of user connections based on the user attribute.

17. The non-transitory machine-readable storage medium of claim 13, wherein the notification includes a message received from the user account.

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